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Description

Identity interface and Identity record

(english translation)

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History

Date	Rev	Author	Changes
19.12.2025	4.1	Alexander Kopp Patrick Hagelkruys	internal review
19.12.2025	4.0	Patrick Hagelkruys	Document translated into English Revising the IdentRecord and the interface. Transferring the schema to an A-Trust namespace Use of 7 gender codes
All entries prior to version 4 refer to an older version of the interface and are presented in a separate document.			

Table 1: Document history

1 Overview

The identity interface is the implementation of the concept „E-Identity Confirmation for BK Activation“ dated October 29, 2009, by Herbert Leitold, Reinhard Posch, and Thomas Rösslerr [HL09].

Based on this concept, A-Trust has implemented a web service that receives and verifies electronically signed and encrypted identity confirmations (IdentRecords). In the next step, a qualified signature product is issued based on these IdentRecords.

This document describes the web service, the schema for an identity record, and the test environment provided by A-Trust.

2 Workflow

The registration authority identifies a person and creates an identity record. This identity record is transmitted to an A-Trust web service.

1. Create an identity record as described in the 3 section
   - Optional: Generate an activation code and unlock PIN
2. Sign the identity record according to the XMLDSig standard [W3C08] with an RO certificate (RO = Registration Officer) from A-Trust
3. Encrypt the identity record as described in the 4 section.
4. Transmit the identity record and the symmetric key to the A-Trust web service, as described in the 5 section.
5. The web service returns the link for redirection to the activation process.

3 Ident record

Example of an ident record:

```
<idr:Confirmation
xmlns:idr="http://www.a-trust.at/namespace/idconfirmation#"
xmlns:dsig="http://www.w3.org/2000/09/xmldsig#"
Version='4'>
  <idr:PersonData>
    <idr:GivenName>Max Xaver</idr:GivenName>
    <idr:FamilyName>Mustermann</idr:FamilyName>
    <idr:SexCode>female</idr:SexCode>
    <idr:DateOfBirth>1976-11-11</idr:DateOfBirth>
    <idr:PlaceOfBirth>Wien</idr:PlaceOfBirth>
    <idr:PhoneNumber>+436505555555</idr:PhoneNumber>
    <idr:EMailAddress>test@test.com</idr:EMailAddress>
    <idr:PostalAddress>
      <idr:AddressLine1>Max Mustermann</idr:AddressLine1>
            ...
      <idr:PostalCode>1000</idr:PostalCode>
    </idr:PostalAddress>
  </idr:PersonData>
  <idr:Identification>
    <idr:IdMethod>IdCard</idr:IdMethod>
    <idr:IdType>PERS</idr:IdType>
    <idr:IdNumber>12345678</idr:IdNumber>
    <idr:IdIssueDate>2011-08-01</idr:IdIssueDate>
    <idr:IdAuthority>Mag. Wien</idr:IdAuthority>
    <idr:IdNation>AT</idr:IdNation>
  </idr:Identification>
  <idr:ProcessData>
    <idr:Name>A-TrustQES</idr:Name>
    <idr:Tag>CustomTag</idr:Tag>
    <idr:Design>A-Trust</idr:Design>
    <idr:Language>DE</idr:Language>
    <idr:Response>
      <idr:Success>https://.../success?token=123455</idr:Success>
      <idr:Error>https://.../error?token=123455</idr:Error>
    </idr:Response>
    <idr:Binding type="MobilePhoneNumber">+436505555555</idr:Binding>
    <idr:HashValue>8jew91qk...1982qw=</idr:HashValue>
  </idr:ProcessData>
  <dsig:Signature>...</dsig:Signature>
</idr:Confirmation>
```

Listing 1: Example ident record

3.1 Structure of the Ident record

The identity record is an XML file with the root element „Confirmation“. The following elements can be defined within it:

- PersonData
- Identification
- ProcessData
- Custom
- Signature

At least the elements `PersonData`, `Identification`, `ProcessData`, and `Signature` must be specified. Likewise, the attribute `Version` must be specified with the value `4`.

3.1.1 PersonData

The element `PersonData` contains the personal data.

- GivenName
- FamilyName
- SexCode (female, male, company, diverse, intersex, open, unknown, not stated)
- DateOfBirth (Format: YYYY-MM-DD)
- PlaceOfBirth
- PhoneNumber
- EMailAddress
- PostalAddress/AddressLine1
- PostalAddress/AddressLine2
- PostalAddress/AddressLine3
- PostalAddress/AddressLine4
- PostalAddress/CountryCode
- PostalAddress/PostalCode

The field `SexCode` can contain the values `male`, `female`, `company`, `diverse`, `intersex`, `open`, `unknown` or `not stated`.

```
<idr:Confirmation
xmlns:idr="http://www.a-trust.at/namespace/idconfirmation#"
xmlns:dsig="http://www.w3.org/2000/09/xmldsig#"
Version='4'>
  <idr:PersonData>
    <idr:GivenName>Max Xaver</idr:GivenName>
    <idr:FamilyName>Mustermann</idr:FamilyName>
    <idr:SexCode>female</idr:SexCode>
    <idr:DateOfBirth>1976-11-11</idr:DateOfBirth>
    <idr:PlaceOfBirth>Wien</idr:PlaceOfBirth>
    <idr:PhoneNumber>+436505555555</idr:PhoneNumber>
    <idr:EMailAddress>test@test.com</idr:EMailAddress>
    <idr:PostalAddress>
      <idr:AddressLine1>Max Mustermann</idr:AddressLine1>
      <idr:AddressLine2></idr:AddressLine2>
      <idr:AddressLine3>Mustergasse 1</idr:AddressLine3>
      <idr:AddressLine4>Wien</idr:AddressLine4>
      <idr:CountryCode>AT</idr:CountryCode>
      <idr:PostalCode>1000</idr:PostalCode>
    </idr:PostalAddress>
  </idr:PersonData>
...
</idr:Confirmation>
```

Listing 2: Example PersonData

AddressLine1 Should contain the name of the person or company. Maximum 32 characters.

AddressLine2 If a company was specified in AddressLine1, the person's name will be entered here. Maximum 32 characters.

AddressLine3 Contains the street name, building number, and apartment number. Maximum 32 characters.

AddressLine4 Contains the city or town. Maximum 24 characters.

CountryCode Country code (2 digits)

PostalCode Postal code (6 digits)

3.1.2 Identification

This field is used to specify identification characteristics. At least the `IDMethod` field must be specified, possible values are:

- AutoIdent
- BankIdent
- EID
- IdCard
- QES
- TrustedDatabase
- VideoIdent

If the person's identification is based on video identification, this must be specified in the `IdMethod` field.

```
<idr:Confirmation
xmlns:idr="http://www.a-trust.at/namespace/idconfirmation#"
xmlns:dsig="http://www.w3.org/2000/09/xmldsig#"
Version='4'>
...
  <idr:Identification>
    <idr:IdMethod>VideoIdent</idr:IdMethod>
  </idr:Identification>
...
</idr:Confirmation>
```

Listing 3: Example IDMethod for video identification

If the signatory is identified by means of an identity document (e.g. by a Registration Officer), the other identity document details must also be provided.

- IdNation
- IdAuthority
- IdIssueDate
- IdNumber
- IdType

The following are the most frequently used values for the ID document type: (*IdType*). A complete list can be found in the interface's XML schema.

REIS Internationaler Reisepass

FUEH Österreichischer Führerschein

IDKA Österreichische Identitätskarte

PERD Deutscher Personalausweis

RA Rechtsanwaltsausweis

eDA eDA Dienstausweis Republik Österreich

eDAO eDA Land Oberösterreich

EDU EDU-Card

GEME Gemeindeausweis

```
<idr:Confirmation
xmlns:idr="http://www.a-trust.at/namespace/idconfirmation#"
xmlns:dsig="http://www.w3.org/2000/09/xmldsig#"
Version='4'>
...
  <idr:Identification>
    <idr:IdMethod>IdCard</idr:IdMethod>
    <idr:IdType>PERS</idr:IdType>
    <idr:IdNumber>12345678</idr:IdNumber>
    <idr:IdIssueDate>2011-08-01</idr:IdIssueDate>
    <idr:IdAuthority>Mag. Wien</idr:IdAuthority>
    <idr:IdNation>AT</idr:IdNation>
  </idr:Identification>
...
</idr:Confirmation>
```

Listing 4: Example ID data

3.1.3 ProcessData

The element `ProcessData` contains information about the required process; elements to be specified:

- Name (A-TrustQES, EuIdentity, WebId, Seal, TestCa, ShortLived)
- Tag
- CustomerName
- Design
- Language (DE, EN)
- Response/Success
- Response/Error
- IdentValidTo
- Binding
- HashValue

```
<idr:Confirmation
xmlns:idr="http://www.a-trust.at/namespace/idconfirmation#"
xmlns:dsig="http://www.w3.org/2000/09/xmldsig#"
Version='4'>
  ...
  <idr:ProcessData>
    <idr:Name>A-TrustQES</idr:Name>
    <idr:Tag>CustomTag</idr:Tag>
    <idr:Design>A-Trust</idr:Design>
    <idr:Language>DE</idr:Language>
    <idr:Response>
      <idr:Success>https://.../success?token=12</idr:Success>
      <idr:Error>https://.../error?token=12</idr:Error>
    </idr:Response>
    <idr:IdentValidTo>2025-01-01</idr:IdentValidTo>
    <idr:Binding
        type="MobilePhoneNumber">+436505555555</idr:Binding>
    <idr:HashValue>8jew91qk...1982qw=</idr:HashValue>
  </idr:ProcessData>
  ...
</idr:Confirmation>
```

Listing 5: Example ProcessData

IdentValidTo The maximum is 30 days, this can be reduced with this parameter.

Response These URLs will be used for redirection after successful activation or in case of an error.

Binding binds the Ident Record to a user factor, possible values for the field type `type` are:

- MobilePhoneNumber
- CINCSN
- ExtCardNumber
- EMail
- ActivationCode

3.1.3.1 Binding und HashValue

Optionally, a `Binding` and `HashValue` can be specified.

These values are passed to the identified person to initiate the issuance of the selected product at a later time via https://www.a-trust.at/Aktivierung.

The `Binding` can be randomly assigned or linked to the person, e.g., their phone number or email address.

The HashValue element contains a Base64-encoded SHA256 hash value derived from the `Binding` and the Freischalte-PIN (Unlock PIN), these are concatenated.

$Hash = Base64(SHA256(UTF8_Bytes(Binding + FreischaltePIN)))$

The Freischalte-PIN is:

- A code generated by the identifying system, which may only be known to the identified person (e.g., activation code sent by registered mail)
- A secret chosen by the identified person and known only to them (e.g., revocation password)

The activation code and Freischalte-PIN values must be UTF-8 encoded. A pseudocode implementation is shown in Listing 6.

```
var identifier = AktivierungsCode+FreischaltePIN;
var utf8bytes = UTF8.GetBytes(identifier);
var hash = SHA256(utf8bytes);
var HashValue = Base64.Encode(hash);
```

Listing 6: Pseudocode for HashValue Generation

3.1.4 Custom

The element `Custom` contains additional fields that are not further defined.

3.1.5 Signature

This element contains an XMLDSIG signature of the identified data.

3.2 Required and recommended elements

The following elements must be specified at a minimum:

- PersonData/GivenName
- PersonData/FamilyName
- PersonData/DateOfBirth
- Identification/IdMethod
- ProcessData/Name
- Signature

4 Encryption of identity record

The signed ident record must be encrypted before being sent to the A-Trust web service. To do this, the ident record is encrypted with an AES 256 key (algorithm: Galois/Counter Mode (GCM) [Wik16a]), and this AES key is then further encrypted with the public RSA key of the A-Trust service (see [Wik16b]). The result of the encryption is two binary blocks: the encrypted identity verification and the encrypted AES 256 key. These data are concatenated, first the encrypted identity verification and then the encrypted AES key.

Process:

1. Generate a new random AES256 key. A separate key must be used for each ident record.
2. Encrypt the signed ident record using AES256/GCM. (UTF-8 representation of the ident record)
   - Algorithm: AES256/GCM
   - IV: Byte array of length 16 filled with 0x00
   - The MAC generated by the GCM algorithm is appended to the encrypted block (default behavior)
3. Encrypting the AES256 key with the public RSA key (RSA-OAEP) of A-Trust
4. The encrypted ident record and the AES encryption are combined. Due to the key length of the RSA key, the data can be separated.

5 Web service for ident records

The following methods are provided for submitting the encrypted identity verification:

POST /v4/Identification: The binary data of the encrypted ident record is transmitted as the POST content. This document is decrypted and checked against the IdConfirmation schema. The signature, the signing time, and the signing certificate are then verified.

GET /v4/Certificate: Returns the certificate for encryption. (Format: Binary)

GET /v4/Certificate/PEM: Returns the certificate for encryption in PEM format.

References

[HL09] Herbert Leitold, Reinhard Posch, Thomas Rössler: *E-Identitätsbestätigung für BK Aktivierung - Konzept*. EGIZ E-Government Innovationszentrum, Oktober 2009.

[W3C08] W3C: *XML Signature Syntax and Processing (Second Edition)*, 2008. https://www.w3.org/TR/xmldsig-core/, besucht: 2017-05-31.

[Wik16a] Wikipedia: *Galois/Counter Mode — Wikipedia, The Free Encyclopedia*, 2016. https://de.wikipedia.org/wiki/Galois/Counter_Mode, besucht: 2017-05-31.

[Wik16b] Wikipedia: *Hybride Verschlüsselung — Wikipedia, The Free Encyclopedia*, 2016. https://de.wikipedia.org/wiki/Hybride_Verschl%C3%BCsselung, besucht: 2017-05-31.